Feature Selection & Model Comparison

MAI511-2 – Advanced Machine Learning | Lab Exercise 3

1. Dataset

Instances
569

Original Features
30

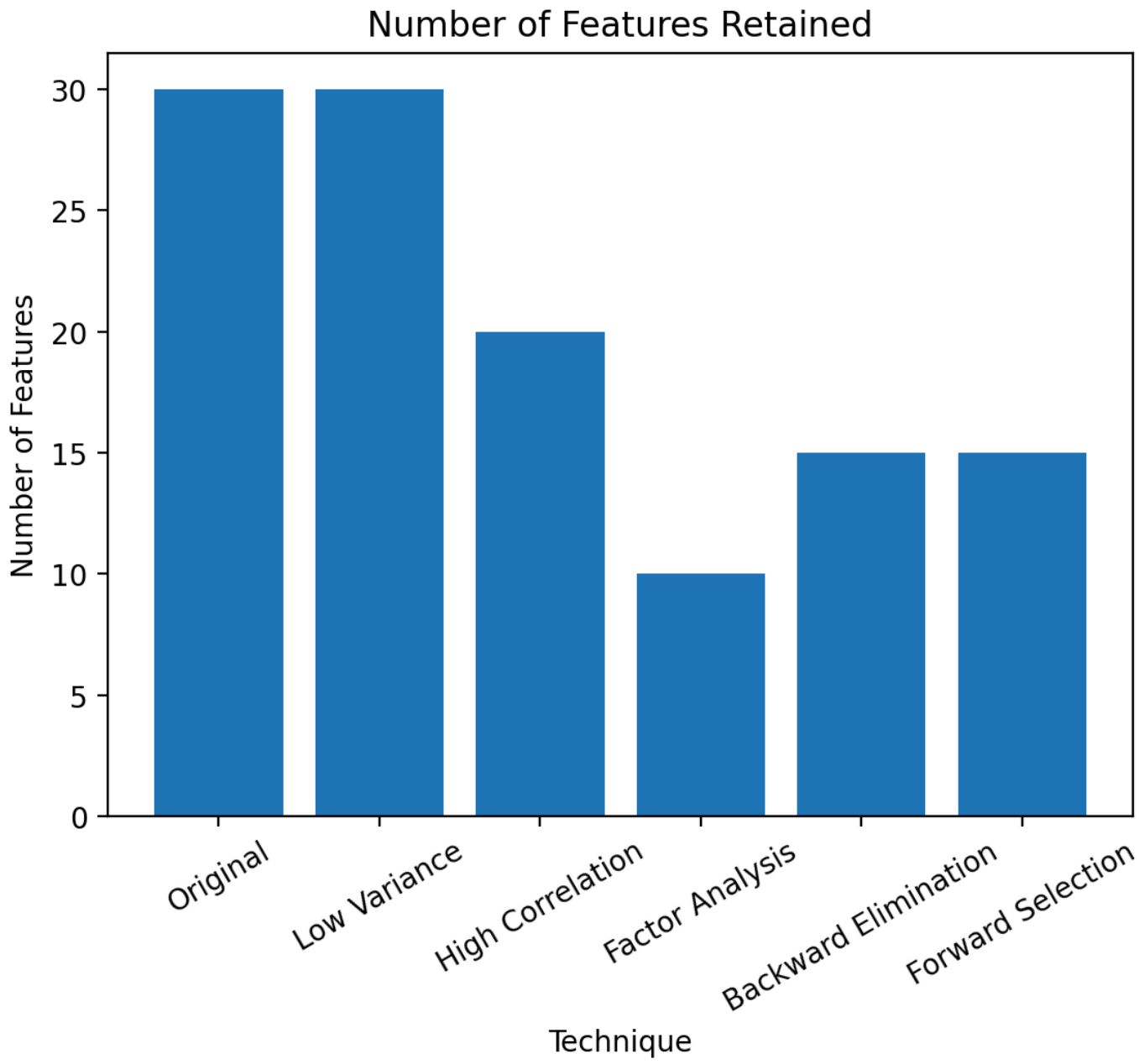
Target Classes
2

Dataset Preview

	id	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness_mean	compactness_mean	concavity_mean	concave points_
0	842302	M	17.99	10.38	122.8	1001	0.1184	0.2776	0.3001	0
1	842517	M	20.57	17.77	132.9	1326	0.0847	0.0786	0.0869	0
2	84300903	M	19.69	21.25	130	1203	0.1096	0.1599	0.1974	0
3	84348301	M	11.42	20.38	77.58	386.1	0.1425	0.2839	0.2414	0
4	84358402	M	20.29	14.34	135.1	1297	0.1003	0.1328	0.198	0
5	843786	M	12.45	15.7	82.57	477.1	0.1278	0.17	0.1578	0
6	844359	M	18.25	19.98	119.6	1040	0.0946	0.109	0.1127	
7	84458202	M	13.71	20.83	90.2	577.9	0.1189	0.1645	0.0937	0
8	844981	M	13	21.82	87.5	519.8	0.1273	0.1932	0.1859	0
9	84501001	M	12.46	24.04	83.97	475.9	0.1186	0.2396	0.2273	0

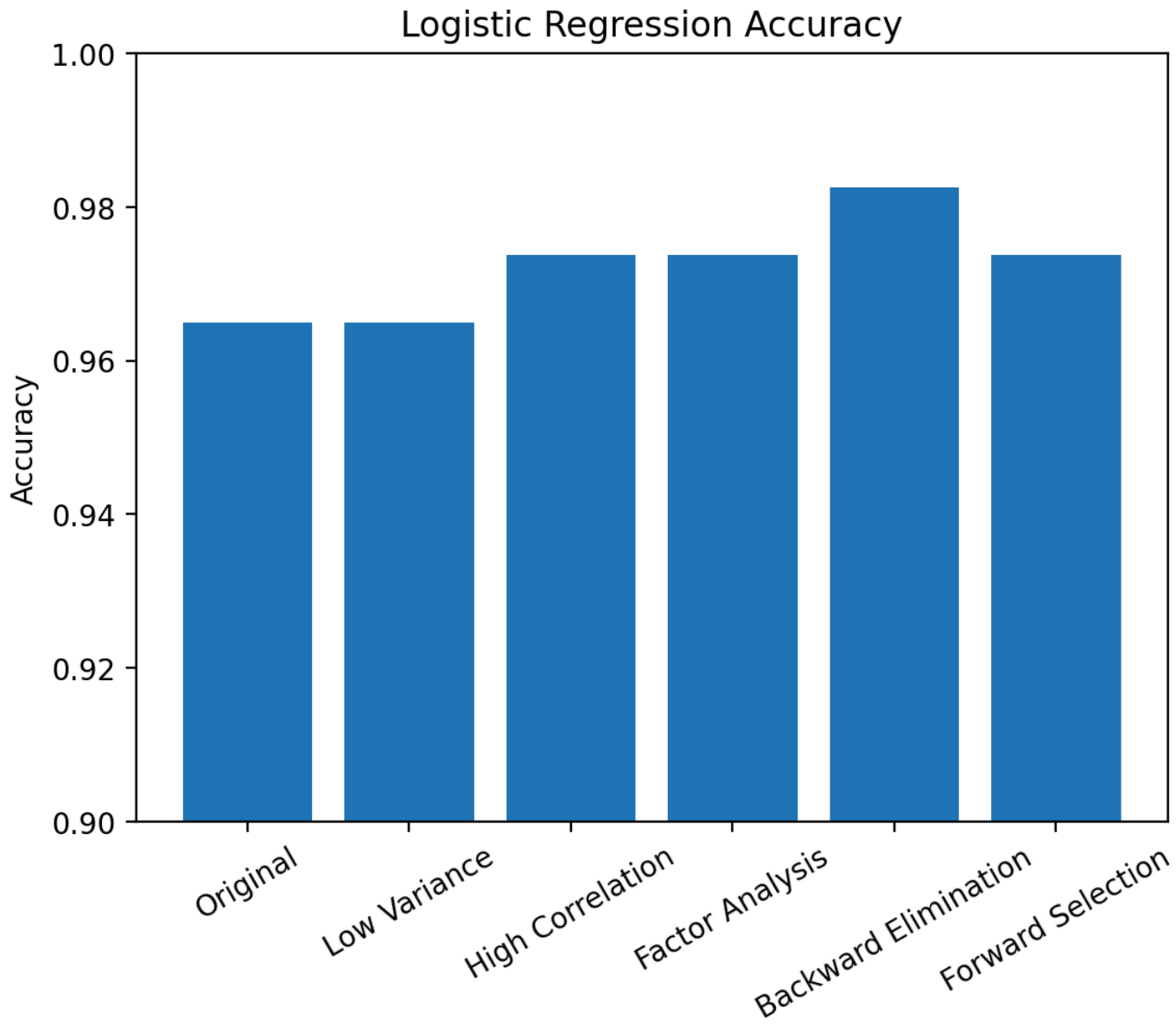
2. Feature Selection Comparison

	Technique	Features
0	Original	
1	Low Variance	
2	High Correlation	
3	Factor Analysis	
4	Backward Elimination	
5	Forward Selection	



3. Logistic Regression Results

	Feature Set	Accuracy	Precision
0	Original	96.49%	9
1	Low Variance	96.49%	9
2	High Correlation	97.37%	10
3	Factor Analysis	97.37%	10
4	Backward Elimination	98.25%	10
5	Forward Selection	97.37%	9



4. Cross-Validation

	Feature Set	Mean CV Accuracy
0	Original	
1	Low Variance	
2	High Correlation	
3	Factor Analysis	
4	Backward Elimination	
5	Forward Selection	

5. GridSearchCV

Linear Regression

Best Parameters:

```
fit_intercept = True
positive = False
```

Test MSE

Logistic Regression

Best Parameters:

```
C = 0.1
solver = liblinear
```

Test Accuracy

0.06265

98.25%

6. Self-Learning Component

The model identifies incorrectly classified training samples and gives them higher weights before retraining.

	Metric	Original Model
0	Accuracy	
1	Precision	
2	Recall	
3	F1 Score	

Self-learning improved accuracy from 96.49% to 97.37%.

7. Observation

Backward Feature Elimination achieved 98.25% test accuracy with only 15 features. The self-learning approach further improved the original model from 96.49% to 97.37% accuracy.

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